

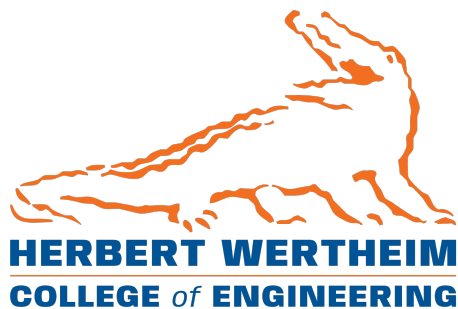
T2: Beta Test Plan

EEL4908 CpE Design 2

Hardened RISC-V Exam Computer

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1 Scope

Beta objective. Two boards accept a “start SEB” command from the server, and SEB launches on both.

Repository: <https://github.com/cannon-spencer/riscv-exam-computer>. Commit tested: *[TODO]*

In scope. Control server, board agent, start and stop scripts, SEB session on the board. *[TODO]* (add or remove items)

Out of scope. *[TODO]*

Notation. *[TODO]* marks something for the team to fill in or confirm; where the blank is not obvious, a short note after the colon says what to write. Nothing marked TODO has been measured or decided.

2 Alpha test results

Describe the results of the alpha testing, how they changed the project, and how they influenced this plan. In the table below, give each test a result (Pass, Fail or Not run) and note the evidence.

2.1 Results against the T1 plan

T1 ID	Result	Evidence and notes
HW-01	<i>[TODO]</i>	<i>[TODO]</i>
HW-02	<i>[TODO]</i>	<i>[TODO]</i>
HW-03	<i>[TODO]</i>	<i>[TODO]</i>
SW-01	<i>[TODO]</i>	<i>[TODO]</i>
INT-01	<i>[TODO]</i>	<i>[TODO]</i>
INT-02	<i>[TODO]</i>	<i>[TODO]</i>
SW-02	<i>[TODO]</i>	<i>[TODO]</i>

Table 1: Alpha results.

2.2 How alpha testing changed the project

[TODO: for each alpha finding, say what you saw and what you changed]

2.3 How alpha testing shaped this plan

[TODO: for each alpha finding, give the beta test ID that checks it]

3 Expected behavior

This section describes behavior as implemented in the source at the tested commit.

3.1 System overview

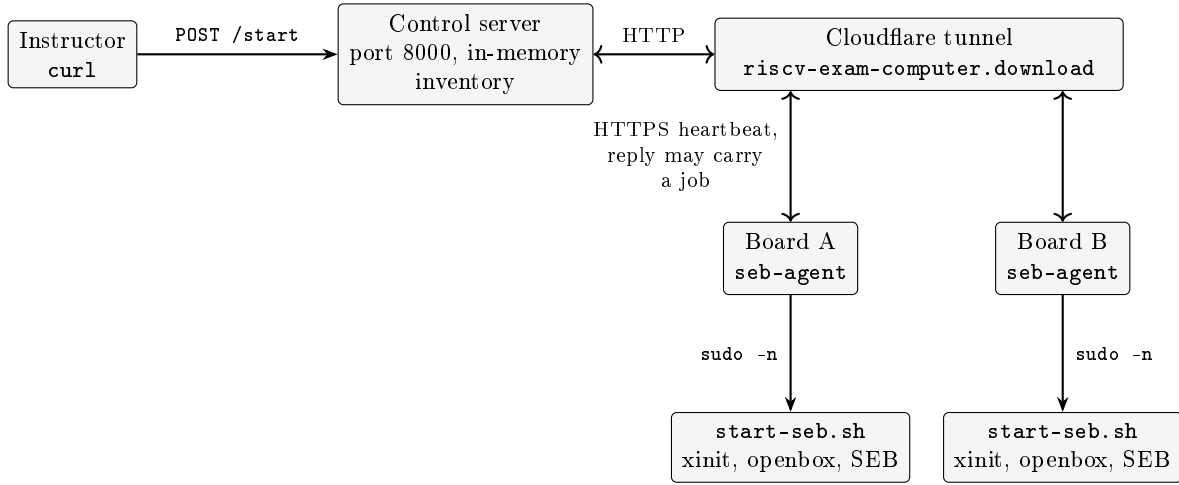


Figure 1: Control path. Boards poll the server; the server never connects to a board.

3.2 Control server

The server (`exam-env/server/src/main.rs`) listens on port 8000 and holds, per host name, the last heartbeat time, the last reported state and at most one pending job (`start` or `stop`).

Route	Expected behavior
POST /heartbeat	Body <code>{"host":..., "state":...}</code> . Records the host, time and state. Replies <code>{"ok":true, "job":...}</code> where <code>job</code> is <code>"start"</code> , <code>"stop"</code> or <code>null</code> . Missing <code>host</code> : 422. Invalid JSON: 400. No JSON content type: 415.
GET /machines	Lists every host seen since the server started, with <code>state</code> , <code>pending</code> and <code>seen_secs_ago</code> , sorted by host.
POST /start	Sets pending <code>start</code> on every host whose last heartbeat is 15 s old or less, and returns the sorted list <code>queued</code> . Returns <code>"queued": []</code> when no host is online.
POST /stop	Same as <code>/start</code> with job <code>stop</code> .

Table 2: Server routes.

Job delivery rules (checked by SRV-01 to SRV-09):

- A job is handed out in the reply to the board's next heartbeat and then cleared.
- A pending `start` is not delivered while the board reports `running`; it stays pending and is delivered the first time the board reports another state.
- A pending `stop` is delivered regardless of reported state.
- There is one pending slot per board, so a later command replaces an earlier undelivered one.
- Inventory is held in memory only.

3.3 Board agent

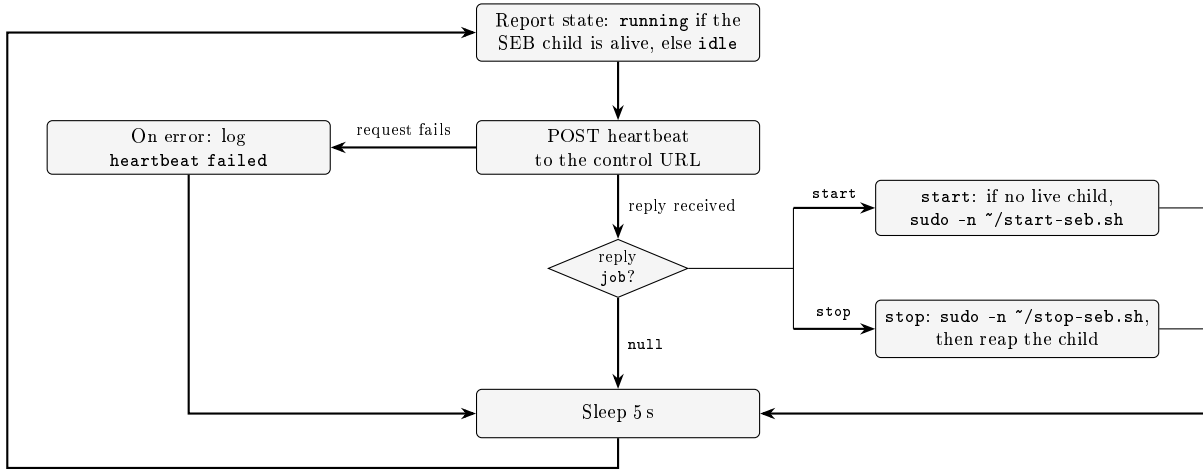


Figure 2: Agent loop (*exam-env/seb-agent/src/main.rs*).

- The agent heartbeats every 5 s (sleep between requests) and reports **running** while the SEB child process it started is alive, otherwise **idle**.
- The reported host name is the `HOSTNAME` environment variable, set by the systemd unit from the `%H` specifier.
- The control URL is compiled into the agent: `https://riscv-exam-computer.download/heartbeat`.
- The agent opens no listening port.
- Sudo for the agent is limited to `/home/orangepi/start-seb.sh` and `/home/orangepi/stop-seb.sh`.

3.4 Board session

Start. `start-seb.sh` runs as root and execs `xinit` on display `:1`, virtual terminal 3, starting `openbox` and then `~/safe-exam-browser --windowed ~/demo-exam.json`.

Stop. `stop-seb.sh` kills the SEB and `xinit` processes and X1 socket holders, rebinds the framebuffer console, restores keyboard mode and switches to virtual terminal 1.

3.5 Quantified requirements

ID	Requirement	Value
R-T1	Heartbeat sleep between requests	5 s (code)
R-T2	A board counts as online while its last heartbeat is no older than	15 s (code)
R-T3	Time from <code>POST /start</code> to SEB running on every online board, in seconds	<i>[TODO]</i>
R-T4	Maximum difference between the two boards' start times, in seconds	<i>[TODO]</i>
R-T5	Time from <code>POST /stop</code> to console restored, in seconds	<i>[TODO]</i>
R-P1	Canvas page load time, in seconds	<i>[TODO]</i>
R-P2	Click-to-response latency in Canvas, in seconds	<i>[TODO]</i>
R-P3	Maximum board temperature during a session, in °C	<i>[TODO]</i>

Table 3: Quantified requirements. Rows marked *TODO* are to be set by the team.

3.6 Not implemented in the tested commit

- The server code has no authentication on any route.
- `/start` and `/stop` address all online boards; there is no per-board targeting.
- The server does not send configuration to boards; the agent starts SEB with a local config file.
- Secure start (the exam opening only under SEB): *[TODO: describe what should happen, and how it would be tested]*
- Other: *[TODO]*

4 Test procedures

4.1 Equipment and conventions

[TODO: list the boards, displays, laptop, network and tools you actually use] Record the commit and board names for every run in Table 10.

4.2 Server (no hardware)

Run `./scripts/test-server-api.sh` against a freshly started local server (`SLOW=1` adds SRV-09). Do not point it at a server that live boards are using.

ID	Procedure and pass criterion
SRV-01	Heartbeat. <code>POST</code> a heartbeat for a new host. Pass: reply contains <code>"ok":true</code> and <code>"job":null</code> .
SRV-02	Inventory. Heartbeat two hosts, then <code>GET /machines</code> . Pass: both hosts listed with state and <code>seen_secs_ago</code> .
SRV-03	Broadcast start. <code>POST /start</code> with both hosts online. Pass: <code>queued</code> lists both; the next heartbeat from a host returns <code>start</code> , the one after returns <code>null</code> .
SRV-04	Start withheld while running. Heartbeat with state <code>running</code> while <code>start</code> is pending. Pass: no job returned and <code>/machines</code> still shows <code>pending:"start"</code> .

ID	Procedure and pass criterion
SRV-05	Stop always delivered. POST /stop, then heartbeat from a running and an idle host. Pass: both receive stop.
SRV-06	Single slot. /start then /stop before the board polls. Pass: only stop is delivered.
SRV-07	Pending start after exit. /start while running, then report idle. Pass: start is delivered.
SRV-08	Bad input. Missing host, invalid JSON, no content type, GET /start. Pass: 422, 400, 415, 405, and /machines still answers 200.
SRV-09	Offline exclusion. Stop heartbeating for 16s, then POST /start. Pass: "queued": [].

Table 4: Server tests.

4.3 Agent and board

ID	Procedure and pass criterion
INS-01	Fresh install. Flash a card, run install-software.sh and install-seb.sh with --host, then any manual steps. Record every manual step needed. Pass: systemctl --user is-active seb-agent prints active, ~/safe-exam-browser exists and is executable, and the board appears in /machines.
AGT-01	Unique identity. On each board run hostname and systemctl --user show seb-agent -p Environment. Pass: /machines lists each board as a separate host.
AGT-02	Cadence. journalctl --user -u seb-agent -o short-precise -n 25. Pass: intervals match R-T1; acceptable range [TODO].
AGT-03	Outage tolerance. Make the control URL unreachable for a set time, then restore it. Pass: the agent keeps running, logs each failed heartbeat, and resumes heartbeating after restoration; duration and recovery time [TODO].
AGT-04	Agent crash recovery. Kill the agent process. Pass: the systemd unit restarts it; time limit [TODO].
AGT-05	Boot without login. Power-cycle a board and do not log in. Pass: the board is listed in /machines; time limit [TODO].
SUDO-01	Sudo scope. As orangepi run sudo -n -l. Pass: only the two scripts are listed.
SES-01	Start from console. Run sudo -n ~/start-seb.sh. Pass: SEB is visible on the display; time limit [TODO].
SES-02	Stop from console. Run sudo -n ~/stop-seb.sh. Pass: no SEB or xinit process remains, the text console is shown and the keyboard works.
SES-03	Repeated cycles. Start and stop repeatedly, recording memory use each cycle. Pass: every cycle meets SES-01 and SES-02; number of cycles and memory tolerance [TODO].
SES-04	Duplicate start. While running, send POST /start twice. Pass: a single SEB session.
SES-05	Config. Validate ~/demo-exam.json and read its start URL. Pass: valid JSON, SEB starts without a password prompt; expected start URL [TODO].

Table 5: Agent and board session tests.

4.4 Integration and demo

ID	Procedure and pass criterion
DEMO-01	Two boards start and stop on one command. Pre: SES-01, SES-02 and AGT-01 passed on both boards; both listed <code>idle</code> in <code>/machines</code> . Steps: (1) note the time; (2) <code>curl -sS -X POST \$CTRL/start</code> ; (3) record when SEB appears on each display; (4) <code>GET /machines</code> ; (5) <code>curl -sS -X POST \$CTRL/stop</code> ; (6) record when each display returns to the console. Number of trials <i>[TODO]</i> . Pass: <code>queued</code> lists both boards; SEB runs on both; <code>/machines</code> shows both <code>running</code> ; after stop both are back at the console. Time limits R-T3 to R-T5 <i>[TODO]</i> .
DEMO-02	Board offline at start. Stop one board's agent, send <code>POST /start</code> . Pass: <code>queued</code> lists only the other board. Restart the agent; record whether it launches SEB. Send <code>/start</code> again and record the result.
DEMO-03	Board loses network while running. With both running, disconnect one board. Record SEB state, the board's <code>seen_secs_ago</code> , and the <code>queued</code> list for a <code>POST /stop</code> . Reconnect and send <code>/stop</code> again.
DEMO-04	Server restart. With both boards running, restart the server. Record how long until both are listed again and with which state.
EDGE-01	Pending start after self-exit. While a board runs, send <code>/start</code> , then end SEB locally. Record whether SEB relaunches. Decide whether that is intended.

Table 6: Integration and demo tests.

4.5 Lockdown and security

ID	Procedure and pass criterion
LD-01	Exit paths. With SEB started by the agent, try the key combinations, mouse actions and window actions the team lists here: <i>[TODO]</i> . Pass: none gives a shell, another application or the console. Record each attempt and outcome.
SEC-01	Secure start. <i>[TODO: write what secure start should mean and how to test it]</i> .
SEC-02	No inbound surface. On a board run <code>ss -ltnp</code> . Pass: no listening socket belongs to <code>seb-agent</code> .
SEC-03	Control-channel access. Record whether <code>POST /start</code> and <code>POST /stop</code> succeed from a machine with no credentials, with no exam running.

Table 7: Lockdown and security tests.

4.6 Performance and endurance

ID	Procedure and pass criterion
PERF-01	Page load. Start SEB with <code>/start</code> ; time the Canvas pages the team lists: <i>[TODO]</i> . Number of runs <i>[TODO]</i> . Report median and maximum. Pass: R-P1.
PERF-02	Input latency. Time click to visible response over a set number of clicks <i>[TODO]</i> . Pass: R-P2.

ID	Procedure and pass criterion
PERF-03	Thermal. Run a Canvas session of set length $[TODO]$ while logging temperature at a set interval $[TODO]$ (log method $[TODO]$). Pass: R-P3.
PERF-04	Endurance. Run SEB for a set duration $[TODO]$. Count crashes, agent restarts and display glitches. Pass: counts and limits $[TODO]$.

Table 8: Performance tests.

4.7 Demo runbook (DEMO-01)

```
# laptop: start server and tunnel (Docker running)
./scripts/deploy-server.sh

# second terminal
CTRL=https://riscv-exam-computer.download
curl -sS $CTRL/machines # both boards listed, state idle
curl -sS -X POST $CTRL/start # queued lists both boards
curl -sS $CTRL/machines # both boards state running
curl -sS -X POST $CTRL/stop # both boards back at the console
```

5 Traceability

Expected behavior	Tests
Server routes and job rules	SRV-01 to SRV-09, EDGE-01
Agent loop, identity, recovery	AGT-01 to AGT-05, SUDO-01, INS-01
Board session start and stop	SES-01 to SES-05
Two-board start and stop (beta objective)	DEMO-01 to DEMO-04
Timing requirements	AGT-02, DEMO-01, SES-01, SES-02
Performance requirements	PERF-01 to PERF-04
Lockdown and security	LD-01, SEC-01 to SEC-03
Alpha findings	<i>[TODO]</i>

Table 9: Expected behavior to tests.

6 Results record

Test	Date	Tester	Boards	Result	Measurements and notes
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Table 10: Results record template.